Assignment - 2

Building Detection from Satellite Imagery

Problem: What are we building and why is it useful?

We are building a YOLO model that takes a satellite image and automatically detects as well as counts the number of buildings in it.

This is useful in the context of our project Pixels-to-Parcels because we are trying to do something similar. From satellite imagery and building detection, we can estimate the number of buildings and houses in an area. This analysis can help in establishing dark stores in the right locations so that we can target a larger market and also maintain fast delivery, which helps in building customer trust.

Data: Where it came from, how many images, what the labels look like?

We used the SpaceNet dataset via Kaggle. It had satellite images of buildings in Paris. There were 1148 train images and corresponding label files.

The labels were provided in YOLO format as .txt files. Each line in a label file represented one building and contained the class ID followed by the normalized bounding box coordinates (x_center, y_center, width, height).

Model: What is YOLOv8, which size did you pick, how many parameters?

YOLOv8 (You Only Look Once, version 8), developed by Ultralytics, is a powerful cutting-edge computer vision model. It is highly efficient and used for tasks like object detection, segmentation and image classification. It is designed to be quite fast and accurate and is the top choice model for modern real-life applications.

The specific model size used is YOLOv8s (Small). For training the model, a total of 11,135,987 parameters were used.

Training: Epochs, batch size, GPU used, training time

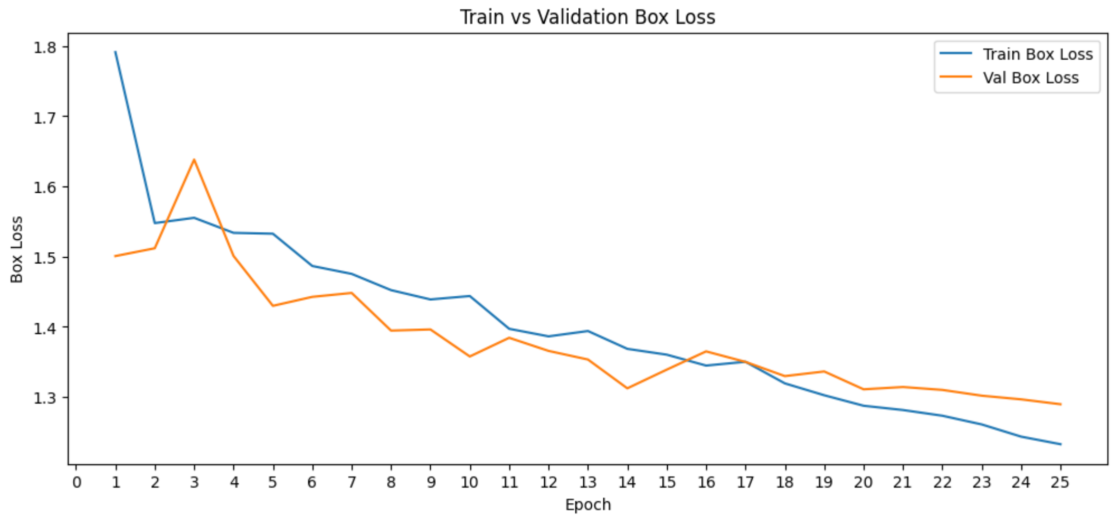
Training was done in 25 Epochs with a batch size of 16. The GPU used to train the model was Tesla T4, 14913MiB and it took total 0.175 hours which is exactly 10 minutes and 30 seconds.

Results: Your metrics table + one training curve image + one sample prediction image

Table:

Metric	Your Value	What it means in plain English
mAP@50	0.821	The model correctly detects buildings with reasonably accurate bounding boxes.
Precision	0.841	Most predicted buildings are actually correct detections.
Recall	0.745	The model successfully finds many buildings but still misses some.
F1 Score	0.790	This represents the balance between precision and recall.

One training curve image:



One sample prediction image:

Observations: What worked well, what the model struggled with, one idea to improve it

The things that worked well were:

• Data Pipeline: The SpaceNet dataset was successfully prepared in YOLOv8 format by creating train, validation, and test splits along with matching label files and a dataset.yaml configuration file.

• Strong Detection Performance: The trained model achieved a Precision of 84.12%, Recall of 74.46%, F1 Score of 0.790, and mAP50 of 0.821 on the test set, indicating that it was able to accurately identify most buildings while maintaining a relatively low number of false detections.

• Fast Predictions: The model processed images quickly, requiring approximately 10.5 ms for inference per image, making it suitable for large-scale satellite image analysis.

The model struggled with:

• Small Buildings: Very small buildings occupied only a few pixels and were sometimes missed by the detector.

• Dense Urban Areas: Closely packed buildings occasionally resulted in missed detections or overlapping bounding boxes.

• Trees, Shadows, and Similar Surfaces: Trees and shadows sometimes covered parts of buildings, and rooftops could look similar to nearby roads or paved areas, making detection more difficult.

• Recall Limitation: Although Precision was high (84.12%), Recall was lower (74.46%), indicating that some buildings were still missed by the model.

One idea to improve the model is to train for more epochs and apply stronger data augmentation techniques. Additional training time combined with augmentations such as Mosaic, flipping, scaling, and brightness adjustments could improve the model's ability to detect small buildings, handle dense urban scenes, and increase Recall while maintaining strong Precision.